

Dataset Overview

Total samples: 44,898

Class distribution:

REAL: 21,417 FAKE: 23,481

The dataset is relatively well balanced, with comparable numbers of real and fake news articles, which helps in building unbiased and reliable classification models.

Text Length Statistics (Before Preprocessing)

Minimum length: 2 words Maximum length: 8,148 words Average length: 417.74 words

Before preprocessing, the news articles show a wide variation in length, ranging from very short textual content to long-form news articles containing extensive details, links, and formatting noise.

Text Length Statistics (After Preprocessing)

Minimum length: 0 words Maximum length: 8,057 words Average length: 411.91 words

After preprocessing, the average text length slightly decreased, reflecting the removal of unnecessary elements such as URLs, special characters, and extra whitespace while retaining the core semantic content required for effective fake news classificatio

Imports

```
import pandas as pd
import numpy as np
import re
import nltk
nltk.download('stopwords')

from nltk.corpus import stopwords
from sklearn.model_selection import train_test_split
from sklearn.feature_extraction.text import TfidfVectorizer
from sklearn.linear_model import LogisticRegression
from sklearn.svm import SVC
from sklearn.ensemble import RandomForestClassifier
from sklearn.metrics import accuracy_score, precision_score, recall_score, f1_score

import tensorflow as tf
from tensorflow.keras.preprocessing.text import Tokenizer
from tensorflow.keras.preprocessing.sequence import pad_sequences
from tensorflow.keras.models import Sequential
from tensorflow.keras.layers import (
    Embedding, RNN, LSTM, GRU, Dense, Dropout,
    Bidirectional, Layer
)
```

```
[nltk_data] Downloading package stopwords to /root/nltk_data...  
[nltk_data] Unzipping corpora/stopwords.zip.
```

2 Load Dataset

```
fake_df = pd.read_csv("/content/drive/MyDrive/Ramesh/Fakenews/Fake.csv")  
true_df = pd.read_csv("/content/drive/MyDrive/Ramesh/Fakenews/True.csv")  
  
fake_df["label"] = 0  
true_df["label"] = 1  
  
df = pd.concat([fake_df, true_df], axis=0)  
df = df.sample(frac=1, random_state=42).reset_index(drop=True)
```

3 Text Preprocessing

```
stop_words = set(stopwords.words("english"))  
  
def clean_text(text):  
    text = text.lower()  
    text = re.sub(r"http\S+|www\S+", "", text)  
    text = re.sub(r"^[a-z\s]", "", text)  
    text = " ".join([w for w in text.split() if w not in stop_words])  
    return text  
  
df["clean_text"] = df["title"].apply(clean_text)
```

4 Train-Test Split

```
X_train, X_test, y_train, y_test = train_test_split(  
    df["clean_text"], df["label"], test_size=0.2, random_state=42  
)
```

5 TF-IDF for ML Models

```
vectorizer = TfidfVectorizer(max_features=5000)  
X_train_vec = vectorizer.fit_transform(X_train)  
X_test_vec = vectorizer.transform(X_test)
```

6 Train ML Models

```
lr = LogisticRegression()  
svm = SVC(probability=True)  
rf = RandomForestClassifier(n_estimators=200)
```

```
lr.fit(X_train_vec, y_train)
svm.fit(X_train_vec, y_train)
rf.fit(X_train_vec, y_train)

lr_prob = lr.predict_proba(X_test_vec)[: ,1]
svm_prob = svm.predict_proba(X_test_vec)[: ,1]
rf_prob = rf.predict_proba(X_test_vec)[: ,1]
```

7 Tokenization for DL Models

```
tokenizer = Tokenizer(num_words=10000)
tokenizer.fit_on_texts(X_train)

X_train_seq = pad_sequences(tokenizer.texts_to_sequences(X_train), maxlen=300)
X_test_seq = pad_sequences(tokenizer.texts_to_sequences(X_test), maxlen=300)
```

8 DL Models (Without Attention)

```
def build_model(model_type):
    model = Sequential([
        Embedding(10000, 128, input_length=300),
        Dropout(0.3)
    ])

    if model_type == "RNN":
        model.add(RNN(tf.keras.layers.SimpleRNNCell(64)))
    elif model_type == "LSTM":
        model.add(LSTM(64))
    elif model_type == "GRU":
        model.add(GRU(64))

    model.add(Dense(1, activation="sigmoid"))
    model.compile(loss="binary_crossentropy", optimizer="adam", metrics=["accuracy"])
    return model
```

```
rnn = build_model("RNN")
lstm = build_model("LSTM")
gru = build_model("GRU")

rnn.fit(X_train_seq, y_train, epochs=3, batch_size=64)
lstm.fit(X_train_seq, y_train, epochs=3, batch_size=64)
gru.fit(X_train_seq, y_train, epochs=3, batch_size=64)

rnn_prob = rnn.predict(X_test_seq).flatten()
lstm_prob = lstm.predict(X_test_seq).flatten()
gru_prob = gru.predict(X_test_seq).flatten()
```

```
/usr/local/lib/python3.12/dist-packages/keras/src/layers/core/embedding.py:97: UserWarning: Argument `input_length` is deprecated. Just remove it.
  warnings.warn(
Epoch 1/3
562/562 — 20s 28ms/step - accuracy: 0.8660 - loss: 0.2930
```

```

Epoch 2/3
562/562 ————— 14s 26ms/step - accuracy: 0.9791 - loss: 0.0603
Epoch 3/3
562/562 ————— 15s 26ms/step - accuracy: 0.9922 - loss: 0.0247
Epoch 1/3
562/562 ————— 15s 15ms/step - accuracy: 0.8663 - loss: 0.3015
Epoch 2/3
562/562 ————— 9s 16ms/step - accuracy: 0.9722 - loss: 0.0755
Epoch 3/3
562/562 ————— 10s 17ms/step - accuracy: 0.9871 - loss: 0.0403
Epoch 1/3
562/562 ————— 11s 16ms/step - accuracy: 0.8599 - loss: 0.2946
Epoch 2/3
562/562 ————— 8s 15ms/step - accuracy: 0.9722 - loss: 0.0748
Epoch 3/3
562/562 ————— 11s 16ms/step - accuracy: 0.9862 - loss: 0.0409
281/281 ————— 2s 7ms/step
281/281 ————— 2s 7ms/step
281/281 ————— 2s 8ms/step

```

9 Attention Layer

```

class Attention(Layer):
    def __init__(self, **kwargs):
        super(Attention, self).__init__(**kwargs)

    def build(self, input_shape):
        self.W = self.add_weight(
            name="att_weight",
            shape=(input_shape[-1], 1),
            initializer="random_normal",
            trainable=True
        )
        self.b = self.add_weight(
            name="att_bias",
            shape=(input_shape[1], 1),
            initializer="zeros",
            trainable=True
        )
        super(Attention, self).build(input_shape)

    def call(self, inputs):
        # inputs shape: (batch, time_steps, features)

        # Compute attention scores
        score = tf.tanh(tf.matmul(inputs, self.W) + self.b)
        score = tf.squeeze(score, axis=-1) # (batch, time_steps)

        # Normalize scores
        attention_weights = tf.nn.softmax(score, axis=1)

        # Weighted sum of inputs
        context = tf.reduce_sum(inputs * tf.expand_dims(attention_weights, -1), axis=-1)

        return context

```

**** 10 Attention-Based DL Models ****

```
def build_attention_model(model_type):
    model = Sequential()
    model.add(Embedding(10000, 128, input_length=300))

    if model_type == "RNN":
        model.add(Bidirectional(RNN(tf.keras.layers.SimpleRNNCell(64), return_sequences=True)))
    elif model_type == "LSTM":
        model.add(Bidirectional(LSTM(64, return_sequences=True)))
    elif model_type == "GRU":
        model.add(Bidirectional(GRU(64, return_sequences=True)))

    model.add(Attention())
    model.add(Dropout(0.3))
    model.add(Dense(1, activation="sigmoid"))

    model.compile(loss="binary_crossentropy", optimizer="adam", metrics=["accuracy"])
    return model
```

```
att_rnn = build_attention_model("RNN")
att_lstm = build_attention_model("LSTM")
att_gru = build_attention_model("GRU")

att_rnn.fit(X_train_seq, y_train, epochs=3, batch_size=64)
att_lstm.fit(X_train_seq, y_train, epochs=3, batch_size=64)
att_gru.fit(X_train_seq, y_train, epochs=3, batch_size=64)

att_rnn_prob = att_rnn.predict(X_test_seq).flatten()
att_lstm_prob = att_lstm.predict(X_test_seq).flatten()
att_gru_prob = att_gru.predict(X_test_seq).flatten()
```

```
Epoch 1/3
562/562 ————— 33s 50ms/step - accuracy: 0.6517 - loss: 0.5880
Epoch 2/3
562/562 ————— 26s 46ms/step - accuracy: 0.9389 - loss: 0.1889
Epoch 3/3
562/562 ————— 26s 46ms/step - accuracy: 0.9459 - loss: 0.1431
Epoch 1/3
562/562 ————— 19s 28ms/step - accuracy: 0.6081 - loss: 0.6116
Epoch 2/3
562/562 ————— 20s 28ms/step - accuracy: 0.9238 - loss: 0.2545
Epoch 3/3
562/562 ————— 15s 27ms/step - accuracy: 0.9543 - loss: 0.1577
Epoch 1/3
562/562 ————— 18s 27ms/step - accuracy: 0.6570 - loss: 0.5287
Epoch 2/3
562/562 ————— 15s 27ms/step - accuracy: 0.9666 - loss: 0.0945
Epoch 3/3
562/562 ————— 15s 26ms/step - accuracy: 0.9858 - loss: 0.0455
281/281 ————— 4s 13ms/step
281/281 ————— 3s 10ms/step
281/281 ————— 4s 13ms/step
```

1 1 Hybrid Function

```
def hybrid(ml, dl):
    return (ml + dl) / 2
```

1 2 Hybrid Models (Without Attention)

```
hybrid_no_att = {
    "RF+RNN": hybrid(rf_prob, rnn_prob),
    "RF+LSTM": hybrid(rf_prob, lstm_prob),
    "RF+GRU": hybrid(rf_prob, gru_prob),

    "LR+RNN": hybrid(lr_prob, rnn_prob),
    "LR+LSTM": hybrid(lr_prob, lstm_prob),
    "LR+GRU": hybrid(lr_prob, gru_prob),

    "SVM+RNN": hybrid(svm_prob, rnn_prob),
    "SVM+LSTM": hybrid(svm_prob, lstm_prob),
    "SVM+GRU": hybrid(svm_prob, gru_prob)
}
```

1 3 Hybrid Models (With Attention)

```
hybrid_att = {
    "RF+AttRNN": hybrid(rf_prob, att_rnn_prob),
    "RF+AttLSTM": hybrid(rf_prob, att_lstm_prob),
    "RF+AttGRU": hybrid(rf_prob, att_gru_prob),

    "LR+AttRNN": hybrid(lr_prob, att_rnn_prob),
    "LR+AttLSTM": hybrid(lr_prob, att_lstm_prob),
    "LR+AttGRU": hybrid(lr_prob, att_gru_prob),

    "SVM+AttRNN": hybrid(svm_prob, att_rnn_prob),
    "SVM+AttLSTM": hybrid(svm_prob, att_lstm_prob),
    "SVM+AttGRU": hybrid(svm_prob, att_gru_prob)
}
```

1 4 Print Hybrid Metrics

```
def print_metrics(model_dict, title):
    rows = []
    for name, probs in model_dict.items():
        preds = (probs > 0.5).astype(int)
        rows.append([
            name,
            accuracy_score(y_test, preds),
            precision_score(y_test, preds),
            recall_score(y_test, preds),
            f1_score(y_test, preds)
        ])
    return rows
```

```
df = pd.DataFrame(rows, columns=["Model", "Accuracy", "Precision", "Recall", "F1"]).round(6)
print(f"\n{title}\n")
print(df.to_string(index=False))
```

```
from sklearn.metrics import accuracy_score, precision_score, recall_score, f1_score
import pandas as pd
```

```
single_models = {
    "Logistic Regression": (lr_prob > 0.5).astype(int),
    "SVM": (svm_prob > 0.5).astype(int),
    "Random Forest": (rf_prob > 0.5).astype(int),
    "RNN": (rnn_prob > 0.5).astype(int),
    "LSTM": (lstm_prob > 0.5).astype(int),
    "GRU": (gru_prob > 0.5).astype(int),
    "AttRNN": (att_rnn_prob > 0.5).astype(int),
    "AttLSTM": (att_lstm_prob > 0.5).astype(int),
    "AttGRU": (att_gru_prob > 0.5).astype(int)
}
```

```
rows = []
```

```
for name, preds in single_models.items():
    rows.append([
        name,
        accuracy_score(y_test, preds),
        precision_score(y_test, preds),
        recall_score(y_test, preds),
        f1_score(y_test, preds)
    ])
```

```
metrics_df = pd.DataFrame(
    rows,
    columns=["Model", "Accuracy", "Precision", "Recall", "F1"]
).round(6)
```

```
print("\nSINGLE MODEL METRICS:\n")
print(metrics_df.to_string(index=False))
```

SINGLE MODEL METRICS:

Model	Accuracy	Precision	Recall	F1
Logistic Regression	0.942428	0.927546	0.953396	0.940293
SVM	0.954677	0.947003	0.958314	0.952625
Random Forest	0.934410	0.930124	0.932084	0.931103
RNN	0.957684	0.950023	0.961593	0.955773
LSTM	0.958575	0.952624	0.960656	0.956623
GRU	0.957572	0.949388	0.962061	0.955682
AttRNN	0.943430	0.939076	0.942155	0.940613
AttLSTM	0.944432	0.943751	0.939110	0.941425
AttGRU	0.960022	0.944735	0.972834	0.958579

```
print_metrics(hybrid_no_att, "HYBRID MODELS (WITHOUT ATTENTION)")
print_metrics(hybrid_att, "HYBRID MODELS (WITH ATTENTION)")
```

HYBRID MODELS (WITHOUT ATTENTION)

Model	Accuracy	Precision	Recall	F1
RF+RNN	0.960468	0.952392	0.965105	0.958707
RF+LSTM	0.965033	0.958933	0.967916	0.963403
RF+GRU	0.964031	0.955038	0.970023	0.962472
LR+RNN	0.959243	0.951434	0.963466	0.957412
LR+LSTM	0.960356	0.954482	0.962529	0.958489
LR+GRU	0.959688	0.950854	0.965105	0.957927
SVM+RNN	0.962027	0.956330	0.964169	0.960233
SVM+LSTM	0.962806	0.957674	0.964403	0.961027
SVM+GRU	0.962806	0.955556	0.966745	0.961118

HYBRID MODELS (WITH ATTENTION)

Model	Accuracy	Precision	Recall	F1
RF+AttRNN	0.954454	0.948640	0.955972	0.952292
RF+AttLSTM	0.948886	0.950793	0.941218	0.945981
RF+AttGRU	0.964699	0.950330	0.976815	0.963391
LR+AttRNN	0.948998	0.942022	0.951288	0.946632
LR+AttLSTM	0.944543	0.943765	0.939344	0.941549
LR+AttGRU	0.960690	0.945823	0.973068	0.959252
SVM+AttRNN	0.955568	0.950640	0.956206	0.953415
SVM+AttLSTM	0.951448	0.951909	0.945667	0.948778
SVM+AttGRU	0.963140	0.950583	0.973068	0.961694

```
!pip install -U transformers accelerate
```

```
Requirement already satisfied: transformers in /usr/local/lib/python3.12/dist-packages (5.0.0)
Collecting transformers
  Downloading transformers-5.2.0-py3-none-any.whl.metadata (32 kB)
Requirement already satisfied: accelerate in /usr/local/lib/python3.12/dist-packages (1.12.0)
Requirement already satisfied: huggingface-hub<2.0,>=1.3.0 in /usr/local/lib/python3.12/dist-packages (from transformers) (1.4.1)
Requirement already satisfied: numpy>=1.17 in /usr/local/lib/python3.12/dist-packages (from transformers) (2.0.2)
Requirement already satisfied: packaging>=20.0 in /usr/local/lib/python3.12/dist-packages (from transformers) (26.0)
Requirement already satisfied: pyyaml>=5.1 in /usr/local/lib/python3.12/dist-packages (from transformers) (6.0.3)
Requirement already satisfied: regex!=2019.12.17 in /usr/local/lib/python3.12/dist-packages (from transformers) (2025.11.3)
Requirement already satisfied: tokenizers<0.23.0,>=0.22.0 in /usr/local/lib/python3.12/dist-packages (from transformers) (0.22.2)
Requirement already satisfied: typer-slim in /usr/local/lib/python3.12/dist-packages (from transformers) (0.24.0)
Requirement already satisfied: safetensors>=0.4.3 in /usr/local/lib/python3.12/dist-packages (from transformers) (0.7.0)
Requirement already satisfied: tqdm>=4.27 in /usr/local/lib/python3.12/dist-packages (from transformers) (4.67.3)
Requirement already satisfied: psutil in /usr/local/lib/python3.12/dist-packages (from accelerate) (5.9.5)
Requirement already satisfied: torch>=2.0.0 in /usr/local/lib/python3.12/dist-packages (from accelerate) (2.10.0+cu128)
Requirement already satisfied: filelock in /usr/local/lib/python3.12/dist-packages (from huggingface-hub<2.0,>=1.3.0->transformers) (3.24.2)
Requirement already satisfied: fsspec>=2023.5.0 in /usr/local/lib/python3.12/dist-packages (from huggingface-hub<2.0,>=1.3.0->transformers) (2025.3.0)
Requirement already satisfied: hf-xet<2.0.0,>=1.2.0 in /usr/local/lib/python3.12/dist-packages (from huggingface-hub<2.0,>=1.3.0->transformers) (1.2.0)
Requirement already satisfied: httpx<1,>=0.23.0 in /usr/local/lib/python3.12/dist-packages (from huggingface-hub<2.0,>=1.3.0->transformers) (0.28.1)
Requirement already satisfied: shellingham in /usr/local/lib/python3.12/dist-packages (from huggingface-hub<2.0,>=1.3.0->transformers) (1.5.4)
Requirement already satisfied: typing-extensions>=4.1.0 in /usr/local/lib/python3.12/dist-packages (from huggingface-hub<2.0,>=1.3.0->transformers) (4.15.0)
Requirement already satisfied: setuptools in /usr/local/lib/python3.12/dist-packages (from torch>=2.0.0->accelerate) (75.2.0)
Requirement already satisfied: sympy>=1.13.3 in /usr/local/lib/python3.12/dist-packages (from torch>=2.0.0->accelerate) (1.14.0)
Requirement already satisfied: networkx>=2.5.1 in /usr/local/lib/python3.12/dist-packages (from torch>=2.0.0->accelerate) (3.6.1)
Requirement already satisfied: jinja2 in /usr/local/lib/python3.12/dist-packages (from torch>=2.0.0->accelerate) (3.1.6)
Requirement already satisfied: cuda-bindings==12.9.4 in /usr/local/lib/python3.12/dist-packages (from torch>=2.0.0->accelerate) (12.9.4)
Requirement already satisfied: nvidia-cuda-nvrtc-cu12==12.8.93 in /usr/local/lib/python3.12/dist-packages (from torch>=2.0.0->accelerate) (12.8.93)
Requirement already satisfied: nvidia-cuda-runtime-cu12==12.8.90 in /usr/local/lib/python3.12/dist-packages (from torch>=2.0.0->accelerate) (12.8.90)
```

```

Requirement already satisfied: nvidia-cuda-cupti-cu12==12.8.90 in /usr/local/lib/python3.12/dist-packages (from torch>=2.0.0->accelerate) (12.8.90)
Requirement already satisfied: nvidia-cudnn-cu12==9.10.2.21 in /usr/local/lib/python3.12/dist-packages (from torch>=2.0.0->accelerate) (9.10.2.21)
Requirement already satisfied: nvidia-cublas-cu12==12.8.4.1 in /usr/local/lib/python3.12/dist-packages (from torch>=2.0.0->accelerate) (12.8.4.1)
Requirement already satisfied: nvidia-cufft-cu12==11.3.3.83 in /usr/local/lib/python3.12/dist-packages (from torch>=2.0.0->accelerate) (11.3.3.83)
Requirement already satisfied: nvidia-curand-cu12==10.3.9.90 in /usr/local/lib/python3.12/dist-packages (from torch>=2.0.0->accelerate) (10.3.9.90)
Requirement already satisfied: nvidia-cusolver-cu12==11.7.3.90 in /usr/local/lib/python3.12/dist-packages (from torch>=2.0.0->accelerate) (11.7.3.90)
Requirement already satisfied: nvidia-cusparse-cu12==12.5.8.93 in /usr/local/lib/python3.12/dist-packages (from torch>=2.0.0->accelerate) (12.5.8.93)
Requirement already satisfied: nvidia-cusparselt-cu12==0.7.1 in /usr/local/lib/python3.12/dist-packages (from torch>=2.0.0->accelerate) (0.7.1)
Requirement already satisfied: nvidia-nccl-cu12==2.27.5 in /usr/local/lib/python3.12/dist-packages (from torch>=2.0.0->accelerate) (2.27.5)
Requirement already satisfied: nvidia-nvshmem-cu12==3.4.5 in /usr/local/lib/python3.12/dist-packages (from torch>=2.0.0->accelerate) (3.4.5)
Requirement already satisfied: nvidia-nvtx-cu12==12.8.90 in /usr/local/lib/python3.12/dist-packages (from torch>=2.0.0->accelerate) (12.8.90)
Requirement already satisfied: nvidia-nvjitlink-cu12==12.8.93 in /usr/local/lib/python3.12/dist-packages (from torch>=2.0.0->accelerate) (12.8.93)
Requirement already satisfied: nvidia-cufile-cu12==1.13.1.3 in /usr/local/lib/python3.12/dist-packages (from torch>=2.0.0->accelerate) (1.13.1.3)
Requirement already satisfied: triton==3.6.0 in /usr/local/lib/python3.12/dist-packages (from torch>=2.0.0->accelerate) (3.6.0)
Requirement already satisfied: cuda-pathfinder~=1.1 in /usr/local/lib/python3.12/dist-packages (from cuda-bindings==12.9.4->torch>=2.0.0->accelerate) (1.3.4)
Requirement already satisfied: typer>=0.24.0 in /usr/local/lib/python3.12/dist-packages (from typer-slim->transformers) (0.24.0)
Requirement already satisfied: anyio in /usr/local/lib/python3.12/dist-packages (from httpx<1,>=0.23.0->huggingface-hub<2.0,>=1.3.0->transformers) (4.12.1)
Requirement already satisfied: certifi in /usr/local/lib/python3.12/dist-packages (from httpx<1,>=0.23.0->huggingface-hub<2.0,>=1.3.0->transformers) (2026.1.4)
Requirement already satisfied: httpcore==1.* in /usr/local/lib/python3.12/dist-packages (from httpx<1,>=0.23.0->huggingface-hub<2.0,>=1.3.0->transformers) (1.0.9)
Requirement already satisfied: idna in /usr/local/lib/python3.12/dist-packages (from httpx<1,>=0.23.0->huggingface-hub<2.0,>=1.3.0->transformers) (3.11)
Requirement already satisfied: h11>=0.16 in /usr/local/lib/python3.12/dist-packages (from httpcore==1.*->httpx<1,>=0.23.0->huggingface-hub<2.0,>=1.3.0->transformers) (0.14.0)
Requirement already satisfied: mpmath<1.4,>=1.1.0 in /usr/local/lib/python3.12/dist-packages (from sympy==1.13.3->torch>=2.0.0->accelerate) (1.3.0)
Requirement already satisfied: click>=8.2.1 in /usr/local/lib/python3.12/dist-packages (from typer>=0.24.0->typer-slim->transformers) (8.3.1)
Requirement already satisfied: rich>=12.3.0 in /usr/local/lib/python3.12/dist-packages (from typer>=0.24.0->typer-slim->transformers) (13.9.4)
Requirement already satisfied: annotated-doc>=0.0.2 in /usr/local/lib/python3.12/dist-packages (from typer>=0.24.0->typer-slim->transformers) (0.0.4)
Requirement already satisfied: MarkupSafe>=2.0 in /usr/local/lib/python3.12/dist-packages (from jinja2->torch>=2.0.0->accelerate) (3.0.3)
Requirement already satisfied: markdown-it-py>=2.2.0 in /usr/local/lib/python3.12/dist-packages (from rich>=12.3.0->typer>=0.24.0->typer-slim->transformers) (4.0.0)
Requirement already satisfied: pygments<3.0.0,>=2.13.0 in /usr/local/lib/python3.12/dist-packages (from rich>=12.3.0->typer>=0.24.0->typer-slim->transformers) (2.19.2)
Requirement already satisfied: mdurl<0.1 in /usr/local/lib/python3.12/dist-packages (from markdown-it-py>=2.2.0->rich>=12.3.0->typer>=0.24.0->typer-slim->transformers) (0.1.2)

```

```

import torch
import numpy as np
import pandas as pd

from transformers import (
    DistilBertTokenizerFast,
    DistilBertForSequenceClassification,
    Trainer,
    TrainingArguments
)

from sklearn.model_selection import train_test_split
from sklearn.metrics import accuracy_score, precision_score, recall_score, f1_score

```

```

# Use title + label (same as earlier ML/DL setup)
df_tr = df[['title', 'label']].copy()
df_tr.dropna(inplace=True)
df_tr.reset_index(drop=True, inplace=True)

X_train, X_test, y_train, y_test = train_test_split(
    df_tr['title'],
    df_tr['label'],
    test_size=0.2,
    random_state=42,
    stratify=df_tr['label']
)

```

)

```
tokenizer = DistilBertTokenizerFast.from_pretrained(
    "distilbert-base-uncased"
)
```

```
train_encodings = tokenizer(
    list(X_train),
    truncation=True,
    padding=True,
    max_length=128
)
```

```
test_encodings = tokenizer(
    list(X_test),
    truncation=True,
    padding=True,
    max_length=128
)
```

/usr/local/lib/python3.12/dist-packages/huggingface_hub/utils/_auth.py:94: UserWarning:

The secret `HF\_TOKEN` does not exist in your Colab secrets.

To authenticate with the Hugging Face Hub, create a token in your settings tab (<https://huggingface.co/settings/tokens>), set it as secret in your Google Colab and restart the notebook. You will be able to reuse this secret in all of your notebooks.

Please note that authentication is recommended but still optional to access public models or datasets.

warnings.warn(

tokenizer_config.json: 100% 48.0/48.0 [00:00<00:00, 3.01kB/s]

vocab.txt: 100% 232k/232k [00:00<00:00, 1.80MB/s]

tokenizer.json: 100% 466k/466k [00:00<00:00, 4.12MB/s]

```
class FakeNewsDataset(torch.utils.data.Dataset):
    def __init__(self, encodings, labels):
        self.encodings = encodings
        self.labels = labels.values

    def __getitem__(self, idx):
        item = {k: torch.tensor(v[idx]) for k, v in self.encodings.items()}
        item["labels"] = torch.tensor(self.labels[idx])
        return item

    def __len__(self):
        return len(self.labels)
```

```
train_dataset = FakeNewsDataset(train_encodings, y_train)
test_dataset = FakeNewsDataset(test_encodings, y_test)
```

```
transformer_model = DistilBertForSequenceClassification.from_pretrained(
    "distilbert-base-uncased",
    num_labels=2
)
```

```
config.json: 100% 483/483 [00:00<00:00, 15.3kB/s]
model.safetensors: 100% 268M/268M [00:02<00:00, 178MB/s]
Loading weights: 100% 100/100 [00:00<00:00, 288.21it/s, Materializing param=distilbert.transformer.layer.5.sa_layer_norm.weight]
```

DistilBertForSequenceClassification LOAD REPORT from: distilbert-base-uncased

Key	Status
-----+-----	-----+-----
vocab_projector.bias	UNEXPECTED
vocab_transform.bias	UNEXPECTED
vocab_transform.weight	UNEXPECTED
vocab_layer_norm.weight	UNEXPECTED
vocab_layer_norm.bias	UNEXPECTED
pre_classifier.weight	MISSING
pre_classifier.bias	MISSING
classifier.bias	MISSING
classifier.weight	MISSING

Notes:

- *UNEXPECTED* :can be ignored when loading from different task/architecture; not ok if you expect identical arch.
- *MISSING* :those params were newly initialized because missing from the checkpoint. Consider training on your downstream task.

```
def compute_metrics(eval_pred):
    logits, labels = eval_pred
    preds = np.argmax(logits, axis=1)

    return {
        "accuracy": accuracy_score(labels, preds),
        "precision": precision_score(labels, preds),
        "recall": recall_score(labels, preds),
        "f1": f1_score(labels, preds)
    }
```

```
training_args = TrainingArguments(
    output_dir="./results",
    eval_strategy="epoch",
    learning_rate=2e-5,
    per_device_train_batch_size=16,
    per_device_eval_batch_size=16,
    num_train_epochs=3,
    weight_decay=0.01,
    logging_dir="./logs",
    logging_steps=100,
    save_strategy="no",
    report_to="none"
)
```

``logging_dir`` is deprecated and will be removed in v5.2. Please set ``TensorBoard_LOGGING_DIR`` instead.

```

trainer = Trainer(
    model=transformer_model,
    args=training_args,
    train_dataset=train_dataset,
    eval_dataset=test_dataset,
    compute_metrics=compute_metrics
)

```

```
trainer.train()
```

 [6735/6735 13:09, Epoch 3/3]

Epoch	Training Loss	Validation Loss	Accuracy	Precision	Recall	F1
1	0.067657	0.045962	0.986414	0.987354	0.984127	0.985738
2	0.027580	0.053369	0.989198	0.987200	0.990196	0.988696
3	0.005226	0.060460	0.989532	0.985628	0.992530	0.989067

TrainOutput(global_step=6735, training_loss=0.041684899049373406, metrics={'train_runtime': 791.2585, 'train_samples_per_second': 136.181, 'train_steps_per_second': 8.512, 'total_flos': 1756357813888776.0, 'train_loss': 0.041684899049373406, 'epoch': 3.0})

```

results = trainer.evaluate()
print(results)

```

 [562/562 00:19]

{'eval_loss': 0.060460448265075684, 'eval_accuracy': 0.989532293986637, 'eval_precision': 0.9856281872971719, 'eval_recall': 0.992530345471522, 'eval_f1': 0.98906722493}

```
bert_results = trainer.evaluate()
```

 [562/562 00:38]

```

final_comparison = pd.DataFrame(
    columns=["Model", "Type", "Accuracy", "Precision", "Recall", "F1-score"]
)

```

```

for _, row in metrics_df.iterrows():
    final_comparison = pd.concat(
        [final_comparison, pd.DataFrame([
            "Model": row["Model"],
            "Type": "ML/DL/Attention",
            "Accuracy": row["Accuracy"],
            "Precision": row["Precision"],
            "Recall": row["Recall"],
            "F1-score": row["F1"]
        ])],
        ignore_index=True
    )

```

/tmp/ipython-input-1812683323.py:2: FutureWarning: The behavior of DataFrame concatenation with empty or all-NA entries is deprecated. In a future version, this will no
 final_comparison = pd.concat(

```

for name, probs in hybrid_no_att.items():
    preds = (probs > 0.5).astype(int)

    final_comparison = pd.concat(
        [final_comparison, pd.DataFrame([
            "Model": name,
            "Type": "Hybrid (No Attention)",
            "Accuracy": accuracy_score(y_test, preds),
            "Precision": precision_score(y_test, preds),
            "Recall": recall_score(y_test, preds),
            "F1-score": f1_score(y_test, preds)
        ])],
        ignore_index=True
    )

```

```

for name, probs in hybrid_att.items():
    preds = (probs > 0.5).astype(int)

    final_comparison = pd.concat(
        [final_comparison, pd.DataFrame([
            "Model": name,
            "Type": "Hybrid (With Attention)",
            "Accuracy": accuracy_score(y_test, preds),
            "Precision": precision_score(y_test, preds),
            "Recall": recall_score(y_test, preds),
            "F1-score": f1_score(y_test, preds)
        ])],
        ignore_index=True
    )

```

```

transformer_row = {
    "Model": "DistilBERT (Fine-tuned)",
    "Type": "Transformer",
    "Accuracy": bert_results["eval_accuracy"],
    "Precision": bert_results["eval_precision"],

```

```
    "Recall": bert_results["eval_recall"],
    "F1-score": bert_results["eval_f1"]
}

final_comparison = pd.concat(
    [final_comparison, pd.DataFrame([transformer_row])],
    ignore_index=True
)
```

```
final_comparison = final_comparison.drop_duplicates(
    subset=["Model", "Type"],
    keep="first"
).reset_index(drop=True)

final_comparison
```

	Model	Type	Accuracy	Precision	Recall	F1-score
0	Logistic Regression	ML/DL/Attention	0.942428	0.927546	0.953396	0.940293
1	SVM	ML/DL/Attention	0.954677	0.947003	0.958314	0.952625
2	Random Forest	ML/DL/Attention	0.934410	0.930124	0.932084	0.931103
3	RNN	ML/DL/Attention	0.957684	0.950023	0.961593	0.955773
4	LSTM	ML/DL/Attention	0.958575	0.952624	0.960656	0.956623
5	GRU	ML/DL/Attention	0.957572	0.949388	0.962061	0.955682
6	AttRNN	ML/DL/Attention	0.943430	0.939076	0.942155	0.940613
7	AttLSTM	ML/DL/Attention	0.944432	0.943751	0.939110	0.941425
8	AttGRU	ML/DL/Attention	0.960022	0.944735	0.972834	0.958579
9	RF+RNN	Hybrid (No Attention)	0.502561	0.478854	0.483660	0.481245
10	RF+LSTM	Hybrid (No Attention)	0.501782	0.477958	0.480859	0.479404
11	RF+GRU	Hybrid (No Attention)	0.501670	0.477980	0.483894	0.480919
12	LR+RNN	Hybrid (No Attention)	0.501559	0.477798	0.482260	0.480019
13	LR+LSTM	Hybrid (No Attention)	0.502227	0.478402	0.480859	0.479627
14	LR+GRU	Hybrid (No Attention)	0.501336	0.477619	0.483193	0.480390
15	SVM+RNN	Hybrid (No Attention)	0.500334	0.476423	0.478758	0.477588
16	SVM+LSTM	Hybrid (No Attention)	0.501559	0.477674	0.479458	0.478565
17	SVM+GRU	Hybrid (No Attention)	0.501336	0.477546	0.481559	0.479544
18	RF+AttRNN	Hybrid (With Attention)	0.500111	0.476179	0.478291	0.477233
19	RF+AttLSTM	Hybrid (With Attention)	0.500780	0.476461	0.470121	0.473270
20	RF+AttGRU	Hybrid (With Attention)	0.501448	0.478013	0.489729	0.483800
21	LR+AttRNN	Hybrid (With Attention)	0.500445	0.476577	0.479692	0.478129
22	LR+AttLSTM	Hybrid (With Attention)	0.500223	0.476000	0.472222	0.474104
23	LR+AttGRU	Hybrid (With Attention)	0.501448	0.478033	0.490196	0.484038
24	SVM+AttRNN	Hybrid (With Attention)	0.499666	0.475669	0.476891	0.476279
25	SVM+AttLSTM	Hybrid (With Attention)	0.502004	0.477841	0.473156	0.475487
26	SVM+AttGRU	Hybrid (With Attention)	0.501670	0.478151	0.487862	0.482958
27	DistilBERT (Fine-tuned)	Transformer	0.989532	0.985628	0.992530	0.989067

```

trainer.save_model("./fake_news_distilbert")
tokenizer.save_pretrained("./fake_news_distilbert")

```

Writing model shards: 100%

1/1 [00:01<00:00, 1.13s/it]

```
('./fake_news_distilbert/tokenizer_config.json',
 './fake_news_distilbert/tokenizer.json')
```

!pip install gradio

```
Requirement already satisfied: gradio in /usr/local/lib/python3.12/dist-packages (5.50.0)
Requirement already satisfied: aiofiles<25.0,>=22.0 in /usr/local/lib/python3.12/dist-packages (from gradio) (24.1.0)
Requirement already satisfied: anyio<5.0,>=3.0 in /usr/local/lib/python3.12/dist-packages (from gradio) (4.12.1)
Requirement already satisfied: brotli>=1.1.0 in /usr/local/lib/python3.12/dist-packages (from gradio) (1.2.0)
Requirement already satisfied: fastapi<1.0,>=0.115.2 in /usr/local/lib/python3.12/dist-packages (from gradio) (0.129.0)
Requirement already satisfied: ffmpy in /usr/local/lib/python3.12/dist-packages (from gradio) (1.0.0)
Requirement already satisfied: gradio-client==1.14.0 in /usr/local/lib/python3.12/dist-packages (from gradio) (1.14.0)
Requirement already satisfied: groovy~=0.1 in /usr/local/lib/python3.12/dist-packages (from gradio) (0.1.2)
Requirement already satisfied: httpx<1.0,>=0.24.1 in /usr/local/lib/python3.12/dist-packages (from gradio) (0.28.1)
Requirement already satisfied: huggingface-hub<2.0,>=0.33.5 in /usr/local/lib/python3.12/dist-packages (from gradio) (1.4.1)
Requirement already satisfied: jinja2<4.0 in /usr/local/lib/python3.12/dist-packages (from gradio) (3.1.6)
Requirement already satisfied: markupsafe<4.0,>=2.0 in /usr/local/lib/python3.12/dist-packages (from gradio) (3.0.3)
Requirement already satisfied: numpy<3.0,>=1.0 in /usr/local/lib/python3.12/dist-packages (from gradio) (2.0.2)
Requirement already satisfied: orjson~=3.0 in /usr/local/lib/python3.12/dist-packages (from gradio) (3.11.7)
Requirement already satisfied: packaging in /usr/local/lib/python3.12/dist-packages (from gradio) (26.0)
Requirement already satisfied: pandas<3.0,>=1.0 in /usr/local/lib/python3.12/dist-packages (from gradio) (2.2.2)
Requirement already satisfied: pillow<12.0,>=8.0 in /usr/local/lib/python3.12/dist-packages (from gradio) (11.3.0)
Requirement already satisfied: pydantic<=2.12.3,>=2.0 in /usr/local/lib/python3.12/dist-packages (from gradio) (2.12.3)
Requirement already satisfied: pydub in /usr/local/lib/python3.12/dist-packages (from gradio) (0.25.1)
Requirement already satisfied: python-multipart=0.0.18 in /usr/local/lib/python3.12/dist-packages (from gradio) (0.0.22)
Requirement already satisfied: pyyaml<7.0,>=5.0 in /usr/local/lib/python3.12/dist-packages (from gradio) (6.0.3)
Requirement already satisfied: ruff>=0.9.3 in /usr/local/lib/python3.12/dist-packages (from gradio) (0.15.1)
Requirement already satisfied: safehttpx<0.2.0,>=0.1.6 in /usr/local/lib/python3.12/dist-packages (from gradio) (0.1.7)
Requirement already satisfied: semantic-version~=2.0 in /usr/local/lib/python3.12/dist-packages (from gradio) (2.10.0)
Requirement already satisfied: starlette<1.0,>=0.40.0 in /usr/local/lib/python3.12/dist-packages (from gradio) (0.52.1)
Requirement already satisfied: tomkit<0.14.0,>=0.12.0 in /usr/local/lib/python3.12/dist-packages (from gradio) (0.13.3)
Requirement already satisfied: typer<1.0,>=0.12 in /usr/local/lib/python3.12/dist-packages (from gradio) (0.24.0)
Requirement already satisfied: typing-extensions~=4.0 in /usr/local/lib/python3.12/dist-packages (from gradio) (4.15.0)
Requirement already satisfied: uvicorn>=0.14.0 in /usr/local/lib/python3.12/dist-packages (from gradio) (0.41.0)
Requirement already satisfied: fsspec in /usr/local/lib/python3.12/dist-packages (from gradio-client==1.14.0->gradio) (2025.3.0)
Requirement already satisfied: websockets<16.0,>=13.0 in /usr/local/lib/python3.12/dist-packages (from gradio-client==1.14.0->gradio) (15.0.1)
Requirement already satisfied: idna>=2.8 in /usr/local/lib/python3.12/dist-packages (from anyio<5.0,>=3.0->gradio) (3.11)
Requirement already satisfied: typing-inspection>=0.4.2 in /usr/local/lib/python3.12/dist-packages (from fastapi<1.0,>=0.115.2->gradio) (0.4.2)
Requirement already satisfied: annotated-doc>=0.0.2 in /usr/local/lib/python3.12/dist-packages (from fastapi<1.0,>=0.115.2->gradio) (0.0.4)
Requirement already satisfied: certifi in /usr/local/lib/python3.12/dist-packages (from httpx<1.0,>=0.24.1->gradio) (2026.1.4)
Requirement already satisfied: httpcore==1.* in /usr/local/lib/python3.12/dist-packages (from httpx<1.0,>=0.24.1->gradio) (1.0.9)
Requirement already satisfied: h11>=0.16 in /usr/local/lib/python3.12/dist-packages (from httpcore==1.*->httpx<1.0,>=0.24.1->gradio) (0.16.0)
Requirement already satisfied: filelock in /usr/local/lib/python3.12/dist-packages (from huggingface-hub<2.0,>=0.33.5->gradio) (3.24.2)
Requirement already satisfied: hf-xet<2.0.0,>=1.2.0 in /usr/local/lib/python3.12/dist-packages (from huggingface-hub<2.0,>=0.33.5->gradio) (1.2.0)
Requirement already satisfied: shellingham in /usr/local/lib/python3.12/dist-packages (from huggingface-hub<2.0,>=0.33.5->gradio) (1.5.4)
Requirement already satisfied: tqdm>=4.42.1 in /usr/local/lib/python3.12/dist-packages (from huggingface-hub<2.0,>=0.33.5->gradio) (4.67.3)
Requirement already satisfied: typer-slim in /usr/local/lib/python3.12/dist-packages (from huggingface-hub<2.0,>=0.33.5->gradio) (0.24.0)
Requirement already satisfied: python-dateutil>=2.8.2 in /usr/local/lib/python3.12/dist-packages (from pandas<3.0,>=1.0->gradio) (2.9.0.post0)
Requirement already satisfied: pytz>=2020.1 in /usr/local/lib/python3.12/dist-packages (from pandas<3.0,>=1.0->gradio) (2025.2)
Requirement already satisfied: tzdata>=2022.7 in /usr/local/lib/python3.12/dist-packages (from pandas<3.0,>=1.0->gradio) (2025.3)
Requirement already satisfied: annotated-types>=0.6.0 in /usr/local/lib/python3.12/dist-packages (from pydantic<=2.12.3,>=2.0->gradio) (0.7.0)
Requirement already satisfied: pydantic-core==2.41.4 in /usr/local/lib/python3.12/dist-packages (from pydantic<=2.12.3,>=2.0->gradio) (2.41.4)
Requirement already satisfied: click>=8.2.1 in /usr/local/lib/python3.12/dist-packages (from typer<1.0,>=0.12->gradio) (8.3.1)
Requirement already satisfied: rich>=12.3.0 in /usr/local/lib/python3.12/dist-packages (from typer<1.0,>=0.12->gradio) (13.9.4)
Requirement already satisfied: six>=1.5 in /usr/local/lib/python3.12/dist-packages (from python-dateutil>=2.8.2->pandas<3.0,>=1.0->gradio) (1.17.0)
Requirement already satisfied: markdown-it-py>=2.2.0 in /usr/local/lib/python3.12/dist-packages (from rich>=12.3.0->typer<1.0,>=0.12->gradio) (4.0.0)
Requirement already satisfied: pygments<3.0.0,>=2.13.0 in /usr/local/lib/python3.12/dist-packages (from rich>=12.3.0->typer<1.0,>=0.12->gradio) (2.19.2)
Requirement already satisfied: mdurl~=0.1 in /usr/local/lib/python3.12/dist-packages (from markdown-it-py>=2.2.0->rich>=12.3.0->typer<1.0,>=0.12->gradio) (0.1.2)
```

```

import gradio as gr
import torch
import torch.nn.functional as F
from transformers import DistilBertTokenizerFast, DistilBertForSequenceClassification

# Load saved model
MODEL_PATH = "./fake_news_distilbert"

tokenizer = DistilBertTokenizerFast.from_pretrained(MODEL_PATH)
model = DistilBertForSequenceClassification.from_pretrained(MODEL_PATH)
model.eval()

# Prediction Function
def predict_news(news_text):

    if news_text.strip() == "":
        return "⚠ Please enter news text.", 0, 0

    inputs = tokenizer(
        news_text,
        return_tensors="pt",
        truncation=True,
        padding=True,
        max_length=128
    )

    with torch.no_grad():
        outputs = model(**inputs)

    probabilities = F.softmax(outputs.logits, dim=1)
    fake_prob = probabilities[0][0].item()
    real_prob = probabilities[0][1].item()

    if real_prob > fake_prob:
        label = "🟢 REAL NEWS"
        confidence = real_prob
    else:
        label = "🔴 FAKE NEWS"
        confidence = fake_prob

    result_text = f"""
    ### 🗣 Prediction Result
    **{label}**

    **Confidence:** {confidence*100:.2f}%
    """

    return result_text, fake_prob, real_prob

# Custom Theme
theme = gr.themes.Soft(
    primary_hue="blue",
    secondary_hue="indigo"
)

```

```
with gr.Blocks(theme=theme, title="Fake News Detection System") as demo:

    gr.Markdown(
        """
        # 📰 Real-World Fake News Detection System
        ### Powered by Fine-Tuned DistilBERT

        Enter any news headline or article below and the model will classify it as **REAL or FAKE**.
        """
    )

    with gr.Row():
        with gr.Column(scale=3):
            news_input = gr.Textbox(
                lines=8,
                placeholder="Paste news headline or full article here...",
                label="📄 Enter News Text"
            )
        predict_btn = gr.Button("🔍 Analyze News", variant="primary")
```